

Measure Theory

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Problem

Consider a random (infinite) string of 0's and 1's

$$(\ell_1, \dots, \ell_n, \dots) \quad \ell_n \in \{0, 1\} \forall n \in \mathbb{N}$$

What is the probability that somewhere along the string, I find 100 consecutive zeros?

Comment

The statement of the problem is unclear. Lebesgue's machinery will clarify this

Define

$$X = \{x = (\ell_1, \dots, \ell_n, \dots) \mid \ell_n \in \{0, 1\} \forall n \in \mathbb{N}\}$$

This is, in a natural way, a compact metric space. Use the "Hamming-style" distance:

For $x = (\ell_1, \dots, \ell_n, \dots)$ and $x' = (\ell'_1, \dots, \ell'_n, \dots)$

$$d(x, x') = \sum_{n \in \mathbb{N}} \frac{|\ell_n - \ell'_n|}{2^n}$$

Since this is a metric space, we can consider the topology:

$$\mathcal{T} = \{G \subseteq X \mid G \text{ is open}\} \subseteq 2^X = \mathcal{P}(X)$$

Can consider

$$\mathcal{B} = \sigma\text{-Alg}(\mathcal{T}) \quad (\text{Borel sigma algebra})$$

Then a probability measure is

$$P : \mathcal{B} \rightarrow [0, 1]$$

Problem 1.1 says:

Let

$$S = \{x = (\ell_1, \dots, \ell_n, \dots) \in X \mid \exists m \in \mathbb{N} \text{ with } \ell_m = \ell_{m+1} = \dots = \ell_{m+99} = 0\}$$

check that $S \in \mathcal{B}$, and compute $P(S)$

1 Algebras of Sets

Fix a non-empty set X .

Definition 1.1**Algebra of Sets**

$\mathcal{A} \subseteq 2^X$ is an *algebra of sets* if

1. $\emptyset \in \mathcal{A}$
2. If $A \in \mathcal{A}$, then $X \setminus A \in \mathcal{A}$
3. If $A, B \in \mathcal{A}$, then $A \cap B \in \mathcal{A}$

Proposition 1.2

Let $\mathcal{A}$ be an algebra of sets on X . Then

1. $X \in \mathcal{A}$
2. If $A, B \in \mathcal{A}$, then $A \cup B \in \mathcal{A}$
3. If $A, B \in \mathcal{A}$, then $A \setminus B \in \mathcal{A}$
4. If $\mathcal{B} \subseteq \mathcal{A}$ is finite, then $\bigcap \mathcal{B} \in \mathcal{A}$
5. If $\mathcal{B} \subseteq \mathcal{A}$ is finite, then $\bigcup \mathcal{B} \in \mathcal{A}$

Example 1.1

Obviously, 2^X and $\{\emptyset, X\}$ are algebras of sets on X

Lemma 1.3

Let $\mathfrak{A}$ be any collection of algebras of sets on X , then $\bigcap \mathfrak{A}$ is an algebra of sets on X .

Proof: First note if $\mathfrak{A} = \emptyset$, then $\bigcap \mathfrak{A} = 2^X$ which is an algebra of sets on X . We now proceed with checking the axioms of an algebra of sets on X .

1. For every $\mathcal{A} \in \mathfrak{A}$, we have $\emptyset \in \mathcal{A}$, thus $\emptyset \in \bigcap \mathfrak{A}$.
2. If $A \in \bigcap \mathfrak{A}$, then $A \in \mathcal{A}$ for every $\mathcal{A} \in \mathfrak{A}$. Thus $X \setminus A \in \mathcal{A}$ for every $\mathcal{A} \in \mathfrak{A}$, and thus $X \setminus A \in \bigcap \mathfrak{A}$.
3. If $A, B \in \bigcap \mathfrak{A}$, then $A, B \in \mathcal{A}$ for every $\mathcal{A} \in \mathfrak{A}$. Thus $A \cap B \in \mathcal{A}$ for every $\mathcal{A} \in \mathfrak{A}$, and thus $A \cap B \in \bigcap \mathfrak{A}$.

□

Proposition 1.4

Let $\mathcal{B} \subseteq 2^X$ be any collection of subsets of X , then there exists a unique algebra of sets $\mathcal{A}$ on X with the following properties:

1. $\mathcal{B} \subseteq \mathcal{A}$
2. Whenever $\mathcal{C}$ is an algebra of sets on X with $\mathcal{B} \subseteq \mathcal{C}$, then we have $\mathcal{A} \subseteq \mathcal{C}$

Proof: Define $\mathfrak{A}$ to be the collection of all algebras of sets on X that contain $\mathcal{B}$. Then $\bigcap \mathfrak{A}$ is an algebra of sets by [Lemma 1.3](#) and satisfies (1) and (2). Uniqueness follows from (2). □

Definition 1.5**Algebra of sets generated by a collection of subsets**

Let $\mathcal{B} \subseteq 2^X$ be any collection of subsets of X . We define the *algebra of sets generated by $\mathcal{B}$* to be the unique algebra from [Proposition 1.4](#). We notate it as $\text{Alg}(\mathcal{B})$.

1.1 Additive set-function on an algebra of sets

Definition 1.6

Let $\mathcal{A} \subseteq 2^X$ be an algebra of sets on X . A function $\mu : \mathcal{A} \rightarrow [0, \infty]$ is called an *additive set-function* on $\mathcal{A}$ if

1. If $A, B \in \mathcal{A}$ with $A \cap B = \emptyset$, then $\mu(A \cup B) = \mu(A) + \mu(B)$
2. $\mu(\emptyset) = 0$

Remark 1.7

The arithmetic of $[0, \infty]$ is standard, except for $0 \cdot \infty = \infty \cdot 0 = 0$, which is a useful convention in measure theory.

Remark 1.8

We almost don't need to require (2), it's just when μ is identically ∞ do things go wrong without (2)

Proposition 1.9

Properties of an additive set function

Let $\mathcal{A}$ be an algebra of sets on X , and let μ be an additive set function on $\mathcal{A}$. Then

1. If $A, C \in \mathcal{A}$ with $A \subseteq C$, then $\mu(A) \leq \mu(C)$
2. If $A, B \in \mathcal{A}$, then $\mu(A \cup B) \leq \mu(A) + \mu(B)$ (sub additivity)
3. If $A_1, \dots, A_n \in \mathcal{A}$ are all pairwise disjoint, then $\mu(A_1 \cup \dots \cup A_n) = \mu(A_1) + \dots + \mu(A_n)$
4. If $A_1, \dots, A_n \in \mathcal{A}$, then $\mu(A_1 \cup \dots \cup A_n) \leq \mu(A_1) + \dots + \mu(A_n)$

2 Sigma Algebras

Definition 2.1

Sigma Algebra

$\mathcal{M} \subseteq 2^X$ is called a *sigma algebra* on X if

1. $\emptyset \in \mathcal{M}$
2. If $A \in \mathcal{M}$, then $X \setminus A \in \mathcal{M}$
3. If $(A_n)_{n=1}^\infty \subseteq \mathcal{M}$, then $\bigcap_{n=1}^\infty A_n \in \mathcal{M}$

Remark 2.2

countable unions work too, and so do finite operations

Lemma 2.3

Let $\mathfrak{A}$ be any collection of sigma algebras on X , then $\bigcap \mathfrak{A}$ is a sigma algebra on X .

Proof: basically identical to the proof for algebras of sets □

Proposition 2.4

Let $\mathcal{B} \subseteq 2^X$ be any collection of subsets of X , then there exists a unique sigma algebra $\mathcal{A}$ on X with the following properties:

1. $\mathcal{B} \subseteq \mathcal{A}$
2. Whenever $\mathcal{C}$ is an algebra of sets on X with $\mathcal{B} \subseteq \mathcal{C}$, then we have $\mathcal{A} \subseteq \mathcal{C}$

Proof: basically identical to the proof for algebras of sets □

Definition 2.5**Sigma algebra generated by a collection of subsets**

Let $\mathcal{B} \subseteq 2^X$ be any collection of subsets of X . We define the *sigma algebra generated by $\mathcal{B}$* to be the unique sigma algebra from Proposition 2.4. We notate it as $\sigma\text{-Alg}(\mathcal{B})$.

Definition 2.6**Borel Sigma Algebra**

Given a topological space $(X, \mathcal{T})$, we call the sigma algebra generated by $\mathcal{T}$ is called the *Borel sigma algebra on X* and written $\mathcal{B}_X := \sigma\text{-Alg}(\mathcal{T})$

2.1 Positive measure on a sigma algebra**Remark 2.7****Increasing limits and series in $[0, \infty]$**

1. Let $(s_n)_{n=1}^\infty$ be an *increasing* sequence in $[0, \infty]$ in the sense that

$$s_1 \leq s_2 \leq \dots$$

Such a sequence is sure to approach a limit $L \in [0, \infty]$. With a slight abuse of notation we can also write the formula $L = \sup\{s_n \mid n \in \mathbb{N}\}$ meaning that L is the smallest upper bound for all the s_n 's. This is usually written in connection to a bounded increasing sequence of numbers in $[0, \infty)$ – our “slight abuse of notation” is that we are also allowing it to cover the case $L = \infty$, which can occur when some of the s_n 's are equal to ∞ , or when $(s_n)_{n=1}^\infty$ is an unbounded sequence in $[0, \infty)$.

2. Let $(t_n)_{n=1}^\infty$ be any sequence in $[0, \infty]$. We can consider the finite sums

$$s_n := \sum_{i=1}^n t_i$$

which forms an increasing sequence in $[0, \infty]$. We can consider $L = \lim_{n \rightarrow \infty} s_n$ as in (1.), and use the familiar notation $L = \sum_{n=1}^\infty t_n$

Definition 2.8**Positive measure**

Let $\mathcal{M} \subseteq 2^X$ be sigma algebra of subsets of X . We say $\mu : \mathcal{M} \rightarrow [0, \infty]$ is a *positive measure* on $\mathcal{M}$ if it satisfies:

1. $\mu(\emptyset) = 0$
- 2.

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n)$$

Whenever $(A_n)_{n=1}^{\infty}$ is a sequence of sets in $\mathcal{M}$ with the property that if $i, j \in \mathbb{N}$ with $i \neq j$ we have $A_i \cap A_j = \emptyset$

Remark 2.9

A positive measure is, in particular, finitely additive. Thus we can use all the properties in [Proposition 1.9](#)

Definition 2.10**Nomenclature involving measures**

1. If $\mathcal{M}$ is a sigma algebra on X , then the pair $(X, \mathcal{M})$ is called a *measurable space*
2. If we also have a positive measure μ on $\mathcal{M}$, then the triple $(X, \mathcal{M}, \mu)$ is called a *measure space*
3. If $\mu(X) < \infty$, then we say μ is a *finite measure*
4. If $\mu(X) = 1$, then we say μ is a *probability measure* and that $(X, \mathcal{M}, \mu)$ is a *probability space*
5. We say that μ is *sigma finite* if there exists a sequence of sets $(X_n)_{n=1}^{\infty}$ in $\mathcal{M}$ with
 - a. $X_1 \subseteq X_2 \subseteq \dots$ (The X_n 's are said to form an increasing chain)
 - b. $\bigcup_{n=1}^{\infty} X_n = X$
 - c. $\mu(X_n) < \infty$ for all $n \in \mathbb{N}$

Proposition 2.11**Sigma sub-additivity**

Let $\mathcal{M}$ be a sigma algebra on X , and let μ be a positive measure on $\mathcal{M}$. If $(A_n)_{n=1}^{\infty}$ is a sequence of sets in $\mathcal{M}$, then

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu(A_n)$$

Proposition 2.12**Continuity along chains**

Let $\mathcal{M}$ be a sigma algebra on X , and let μ be a positive measure on $\mathcal{M}$.

1. If $(A_n)_{n=1}^{\infty}$ is an increasing chain in $\mathcal{M}$, then

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} \mu(A_n)$$

2. If $(A_n)_{n=1}^{\infty}$ is a decreasing chain in $\mathcal{M}$ with $\mu(A_1) < \infty$, then

$$\mu\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} \mu(A_n)$$